



From lecture 17: We learned about oscillatory motions in the ocean, called waves and had a taste of a real-physical oceanography research activity on wind-driven currents. Today we will continue studying the ocean circulation. However, we will focus on water moving near the coasts

Lecture Session 18

1. Coastal shape and environments;
2. Examples of depositional/erosion systems;
3. Estuaries

Icebreaker discussion

1. Organize yourself into groups of 5.
2. Identify three unique coastal features.
3. Discuss ways to categorize the coastal features into groups based on similarities.



An aerial photograph of a tropical coastline. A narrow, white sandy beach runs diagonally across the frame. To the left of the beach is the deep blue ocean. To the right is a shallow lagoon with turquoise water, showing intricate patterns of sandbars and coral reefs. The text "Shaping Coasts" is overlaid in white, centered horizontally and slightly above the beach line.

Shaping Coasts

Coasts Are Shaped by Marine and Terrestrial Processes

- Coast – zone affected by coastal processes
 - Shore – ocean meets land
 - Location – depends on global tectonic activity and volume of ocean water
 - Shape – depends on uplift, subsidence
 - Erosion, and redistribution of sediment

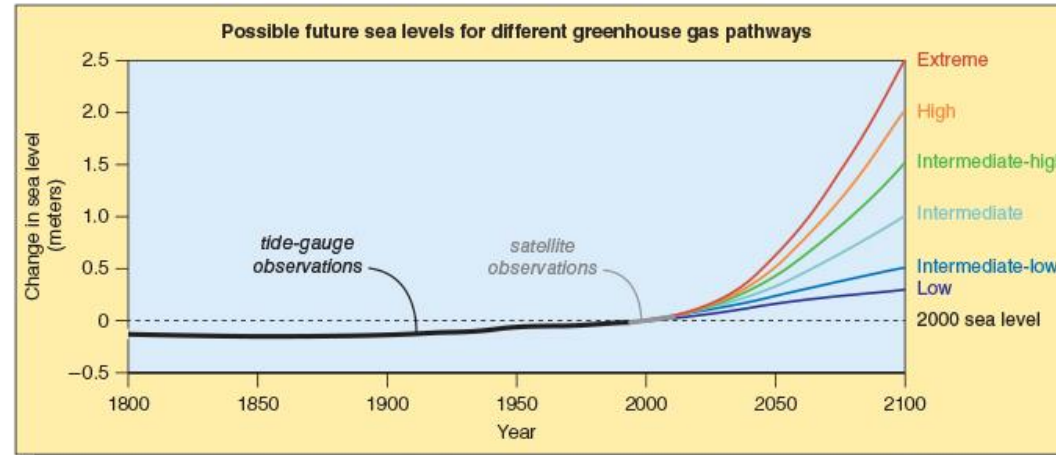
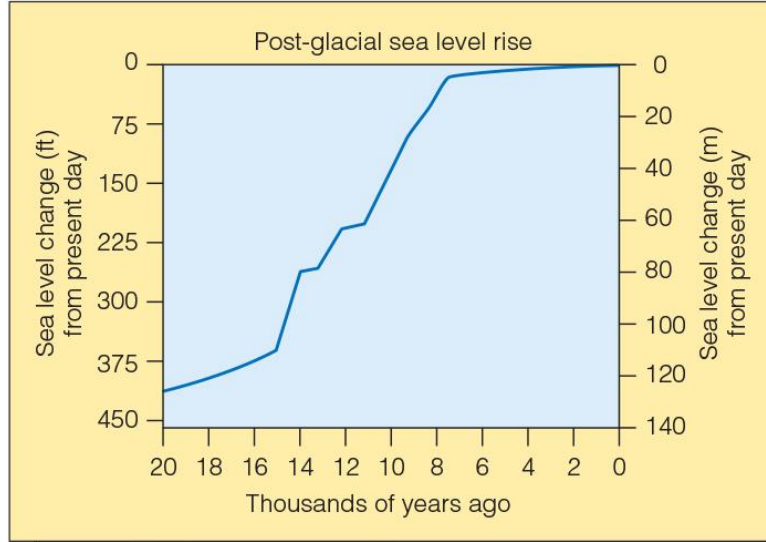


Coasts Are Shaped by Marine and Terrestrial Processes

- Change of sea level

- Three factors cause eustatic change (water level)
 - Amount of water in ocean
 - Volume of Earth's ocean basins
 - Water temperature
- Local changes in sea level
 - Tectonic and isostatic factors (vertical motions of the ocean bottom)
 - Wind currents, seiches, storm surges, climate variability, or other water in motion

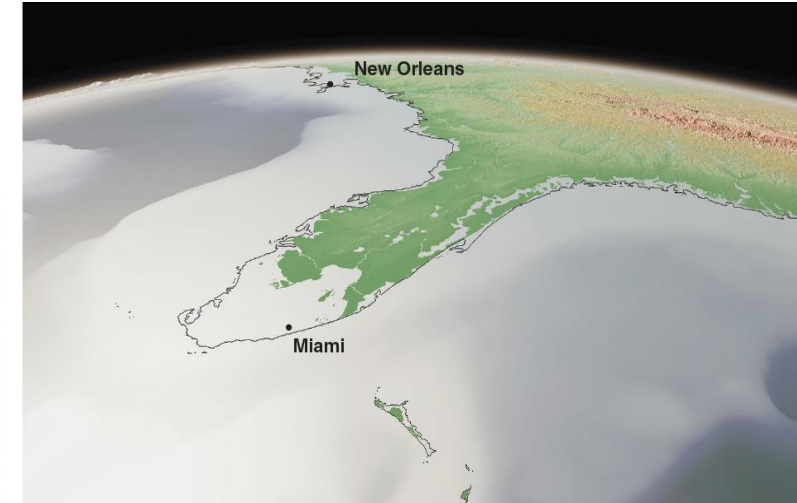
Coasts Are Shaped by Marine and Terrestrial Processes



Source: U.S. GLOBAL CHANGE RESEARCH PROGRAM

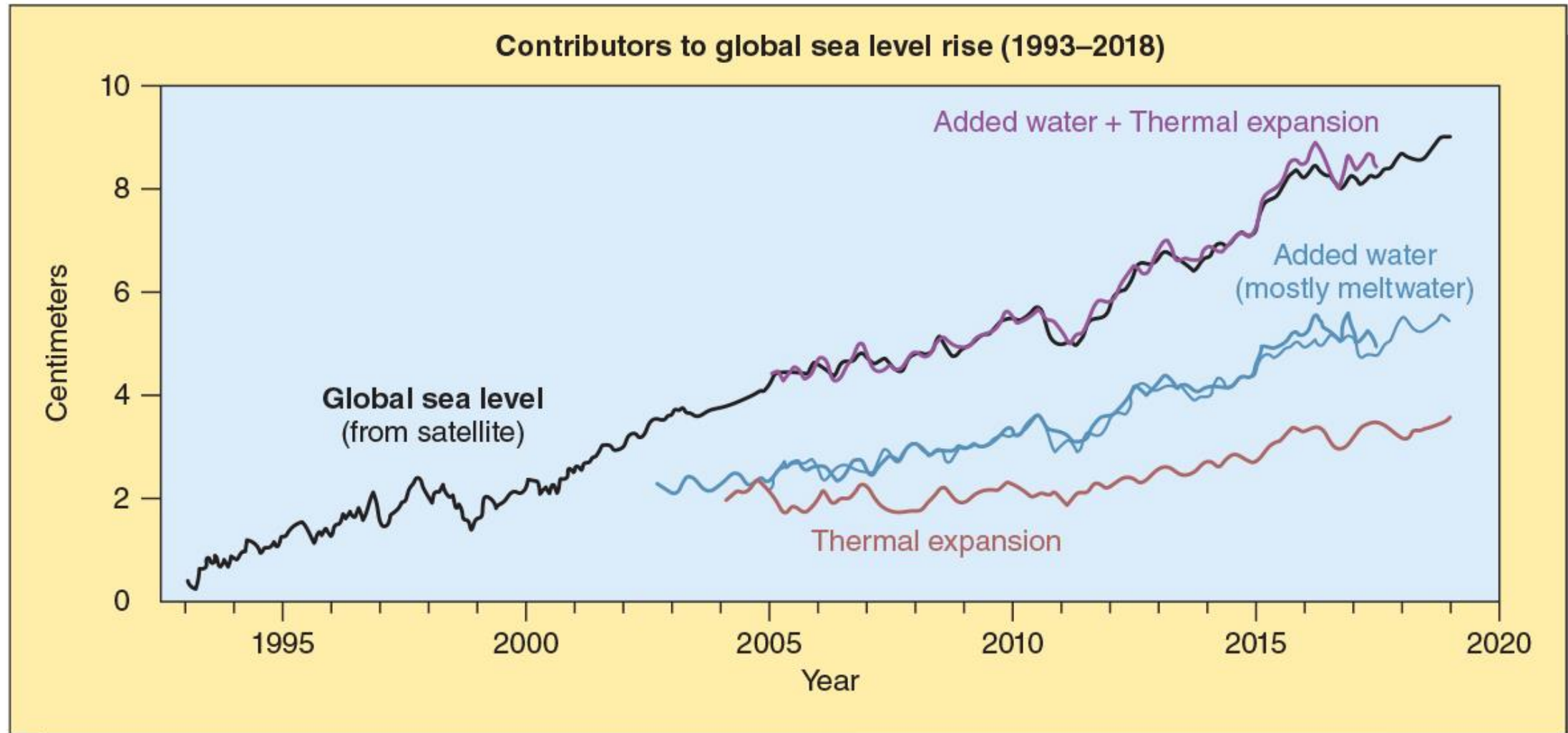


William Haxby/Lamont-Doherty Earth Observatory of Columbia



William Haxby/Lamont-Doherty Earth Observatory of Columbia

Coasts Are Shaped by Marine and Terrestrial Processes



An aerial photograph of a coastal area. The top half of the image shows deep blue ocean water. Below it is a narrow, light-colored sandy beach. Further out, the water turns a vibrant turquoise color, indicating a shallow reef flat or lagoon. The bottom half of the image shows a vast, flat, greenish-brown landscape, likely a coastal plain or wetland. The text 'Rocky coasts' is overlaid in white on the dark blue water.

Rocky coasts

- High-energy coast – rapid erosion
- Complex features
 - Sea cliffs
 - Sea caves
 - Wave-cut platforms

Results of Wave Action On a Coast

David A. Litman/Shutterstock.com



January 2nd



Jeff Hansen, USGS

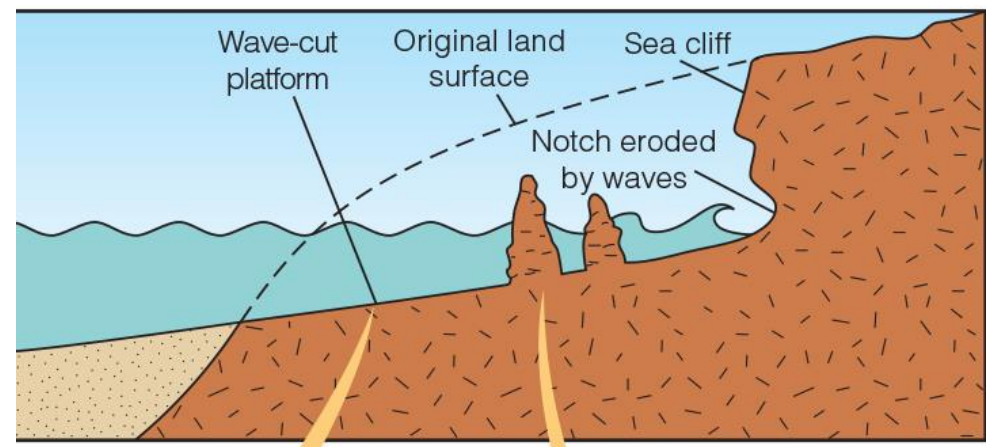
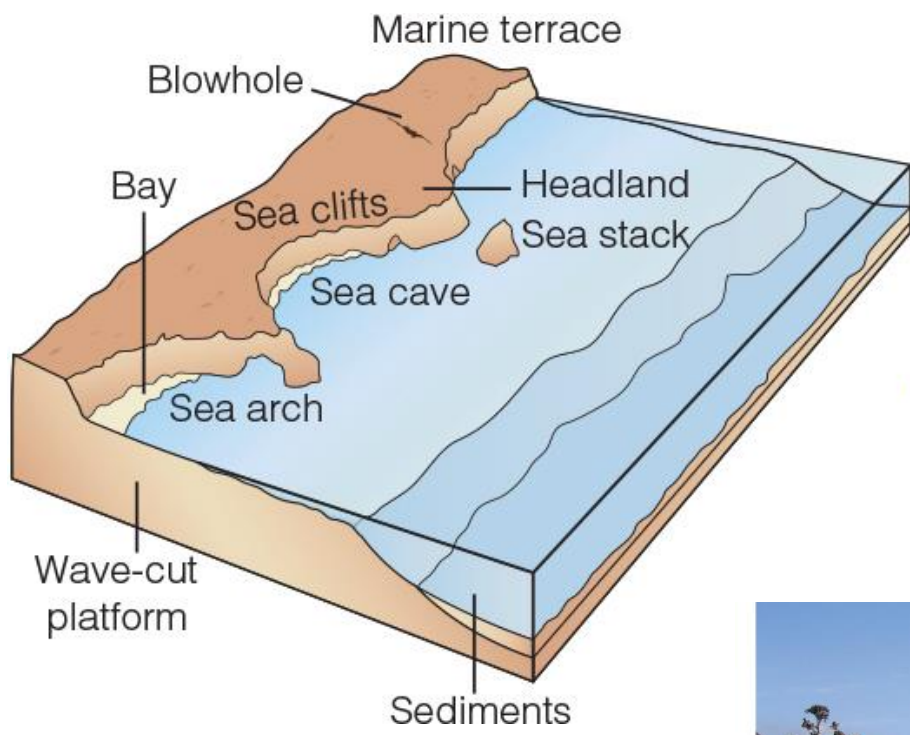
January 20th



January 24th

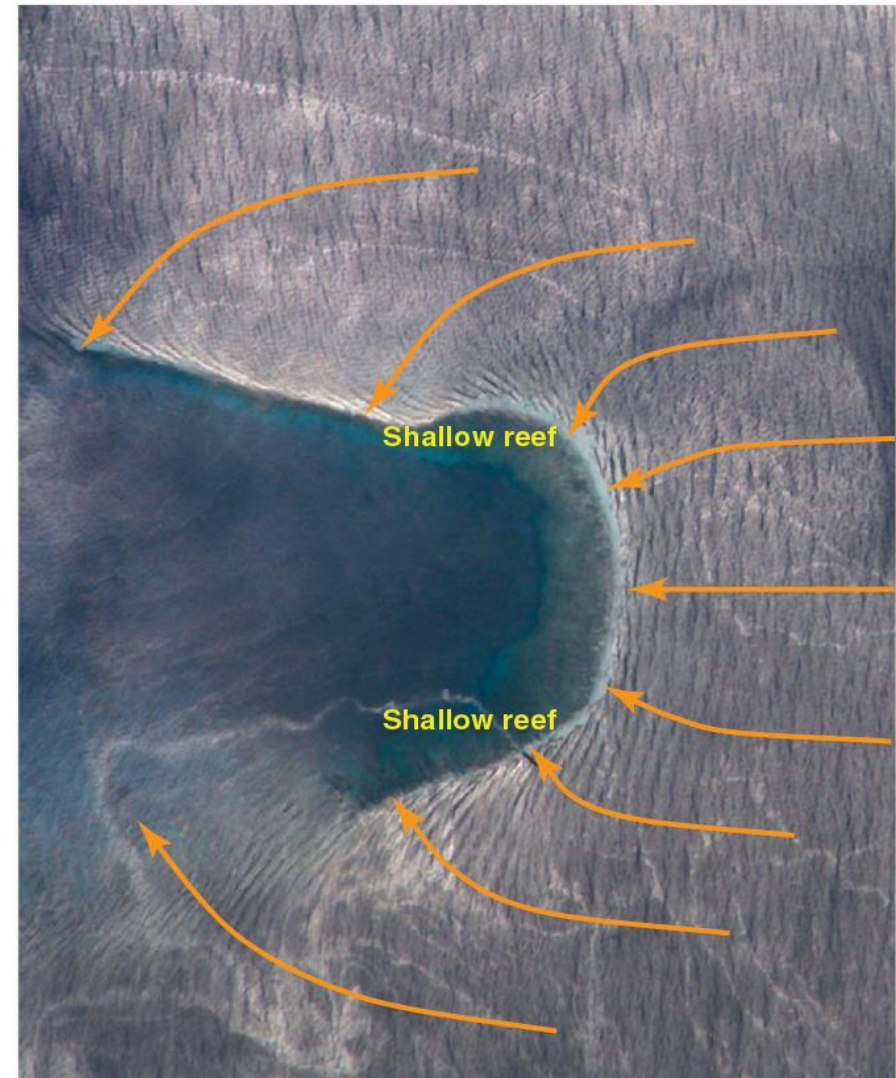
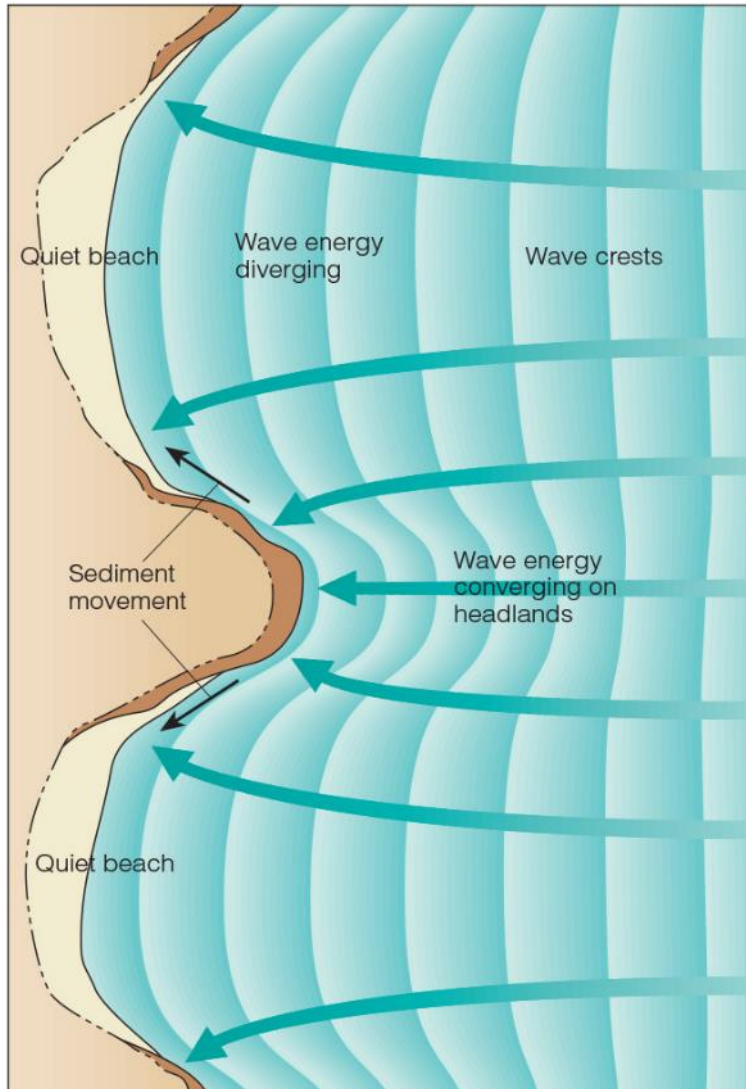


Results of Wave Action On a Coast



Jeff Hansen, USGS; CSLD/Shutterstock.com

Results of Wave Action On a Coast



Volcanism and Tectonism



Dorian Weisel/Corbis



Tom Garrison

An aerial photograph of a coastal area. The top half of the image shows deep blue ocean water. Below the water is a narrow, light-colored sandy beach. Further inland, there is a large, flat, light-colored area, likely a salt flat or a dry lagoon. The text "Depositional Coasts" is overlaid in white, centered horizontally across the middle of the image.

Depositional Coasts

Beaches Dominate Depositional Coasts

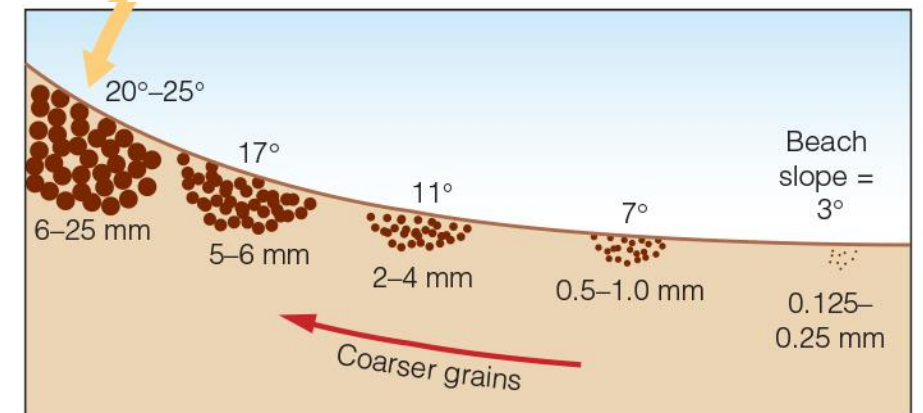
- Beach – loose particles
- Wave action, particle size, and permeability
 - Swash – carries particles onshore
 - Backwash – carries particles offshore



Beaches Dominate Depositional Coasts

- Beach – loose particles
- Wave action, particle size, and permeability
 - Swash – carries particles onshore
 - Backwash – carries particles offshore

Particle size will result in different beaches' profiles

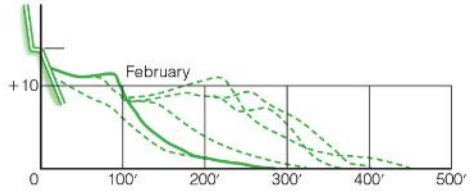
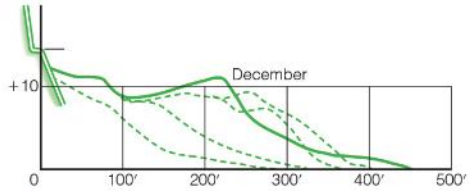
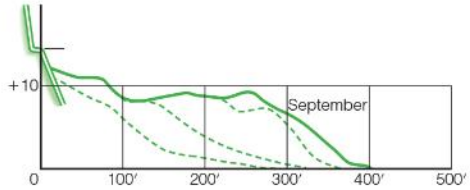
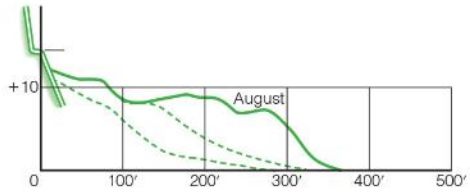
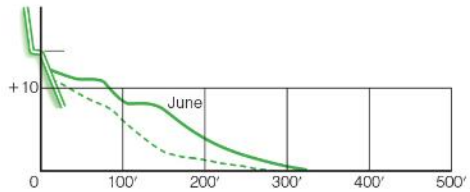
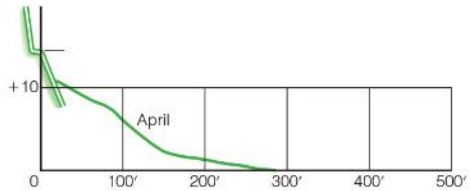


Example: Seasonal Beach Changes

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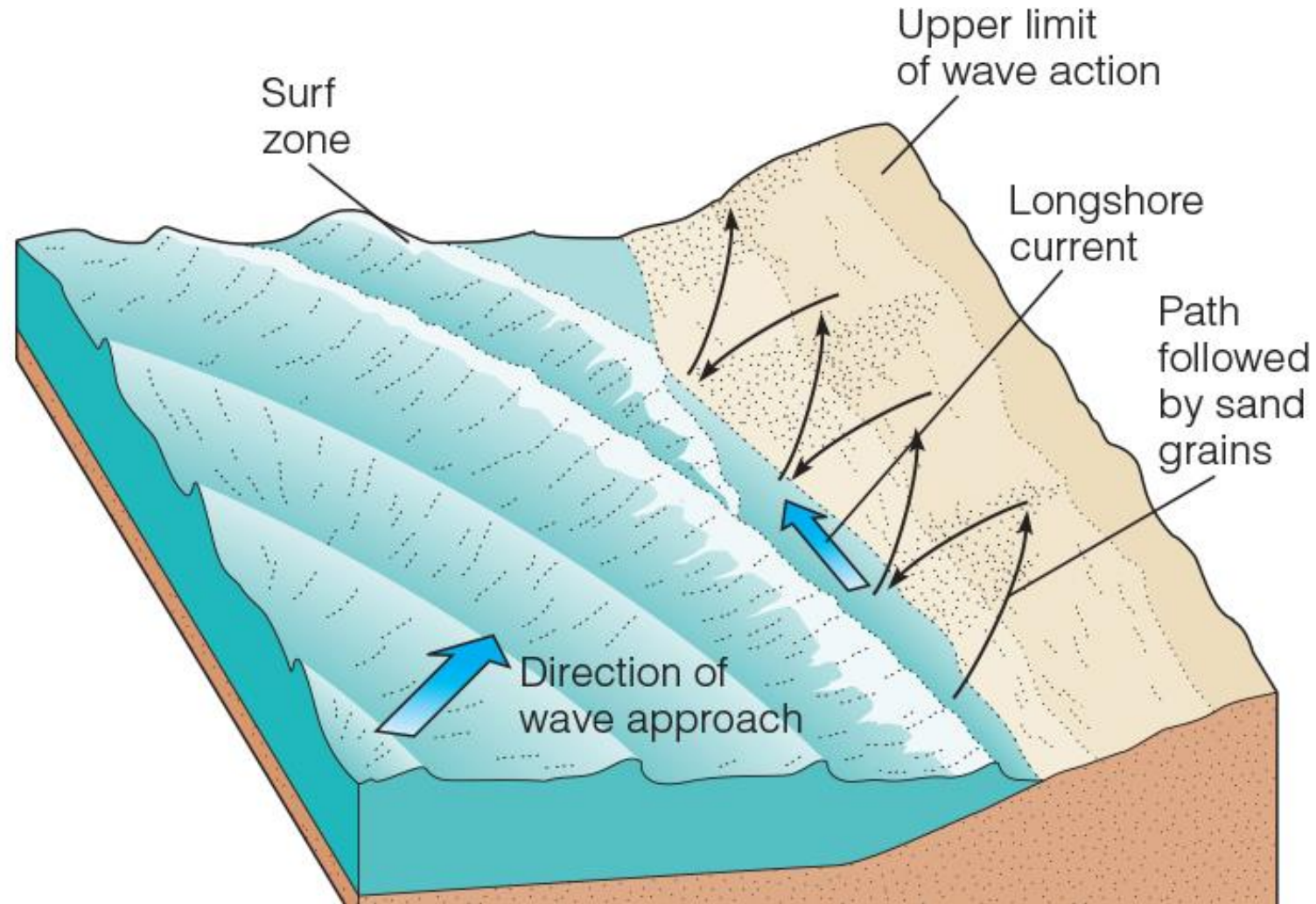


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Waves Transport Beach Sediment

- Longshore drift and longshore current



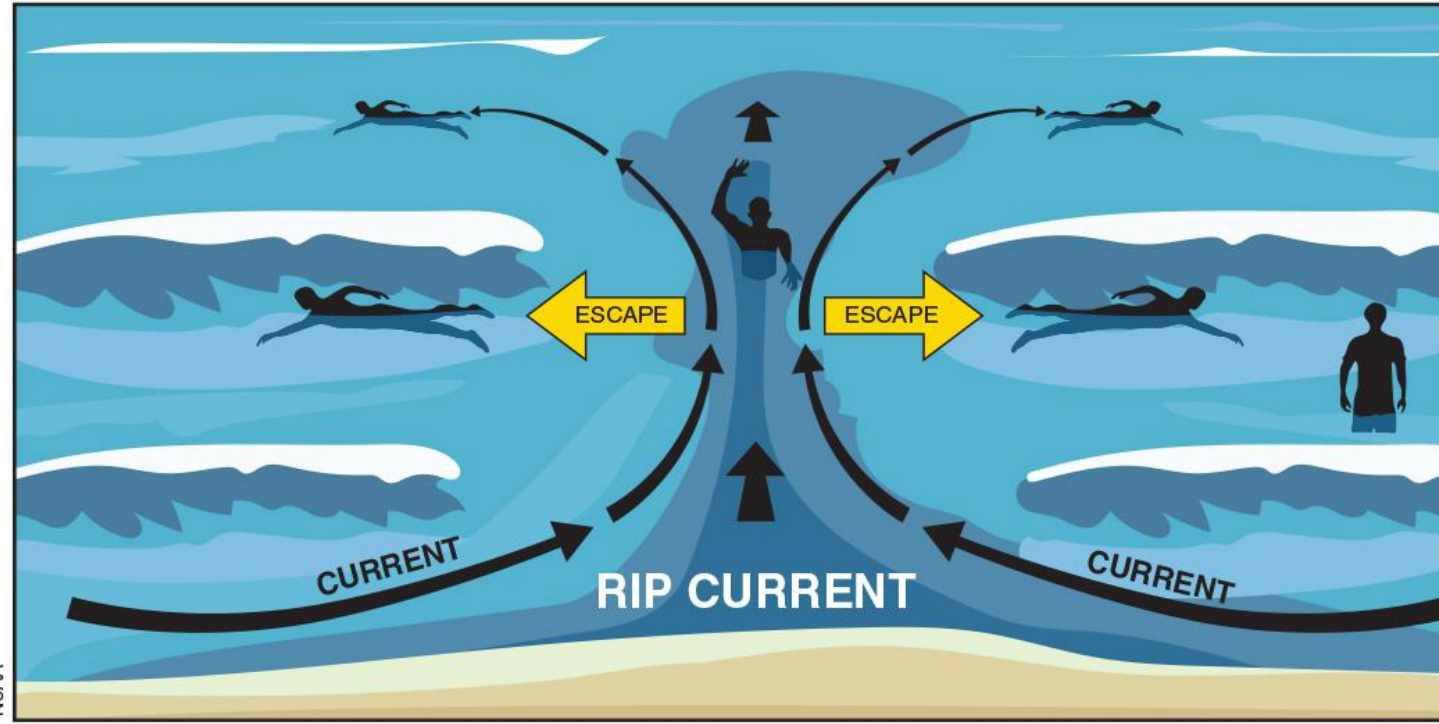
Waves Transport Beach Sediment

- Longshore drift and longshore current



NOAA

NOAA



Examples of Larger-Scale Features Accumulate on Depositional Coasts – Beaches' are not the only depositional environment

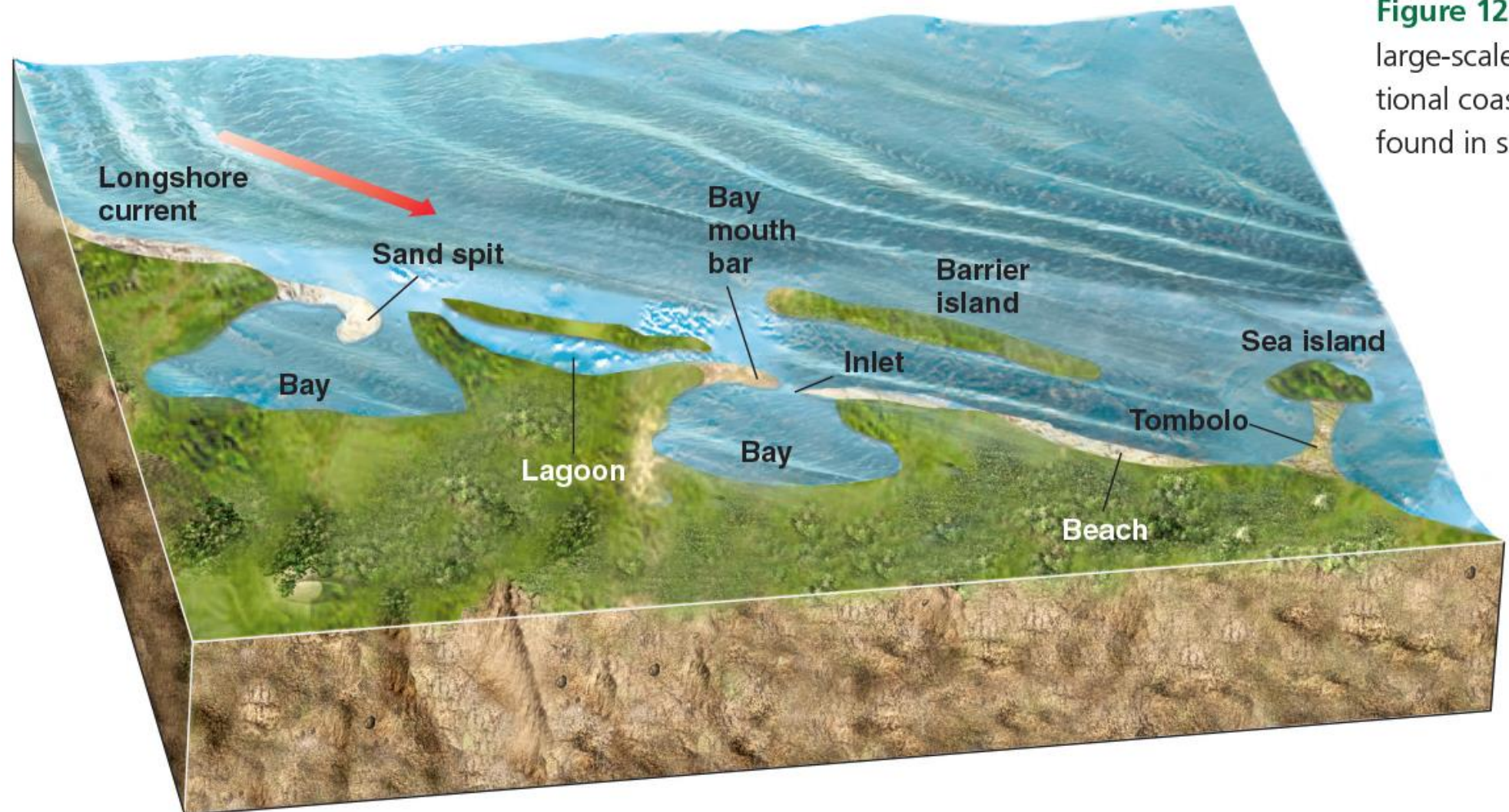
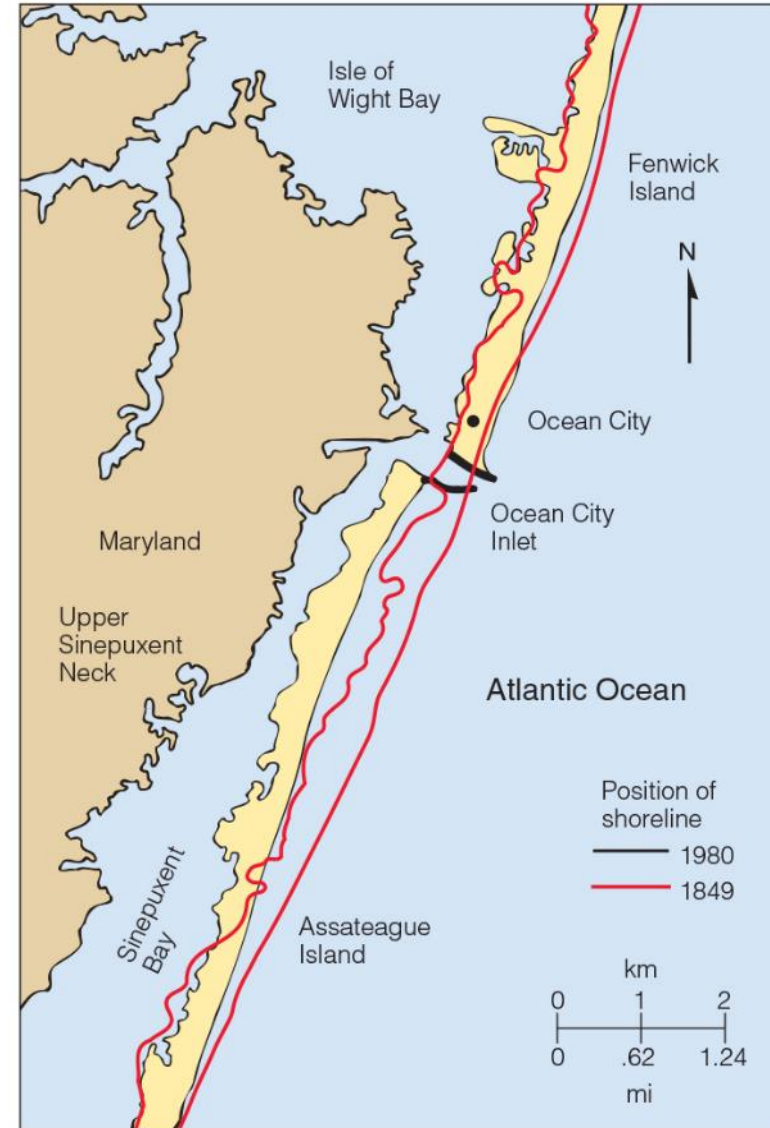
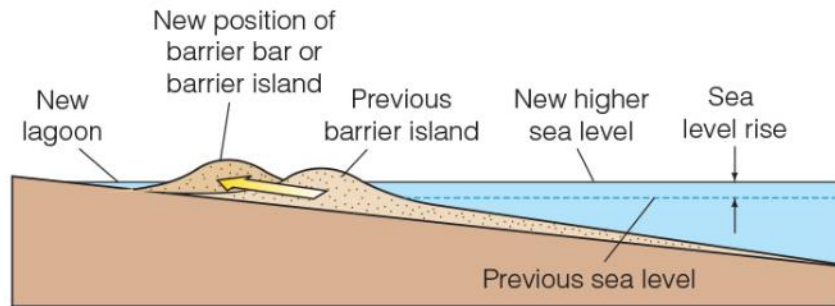


Figure 12.
large-scale
tional coas
found in su

Barrier Islands and Deltas

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Barrier Islands and Deltas



W. Demetras, O. C. Camera



Mark Wolfe/FEMA News Photo



Mark Wolfe/FEMA News Photo

Barrier Islands and Deltas



U.S. Geological Survey





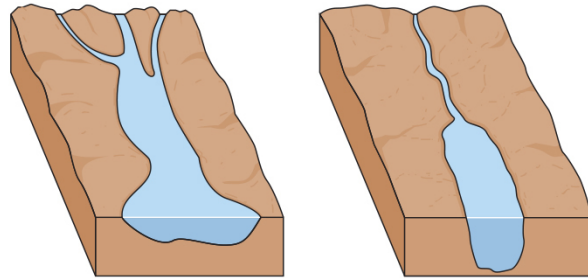
Estuaries

In summary: Underwater environment present on coastal areas
where freshwater meets the ocean

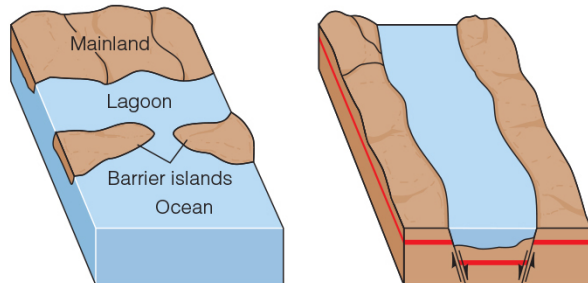
Freshwater Meets the Ocean in Estuaries

- Mixing fresh and salt water
- Estuaries classified by origin
- Characteristics influenced by water density and flow

a Drowned river mouths: the mouths of the James, York, and Susquehanna rivers; Chesapeake Bay; Sydney Harbour, Australia.

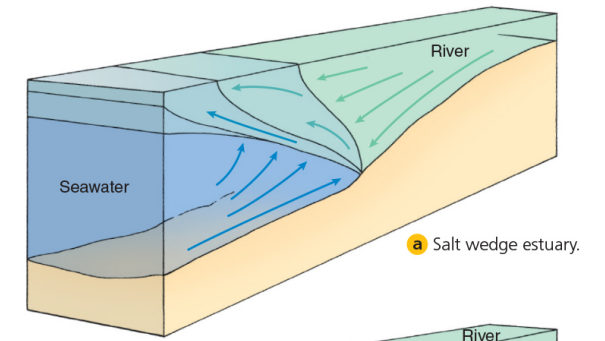


c Bar-built: Albemarle and Pamlico sounds in North Carolina.

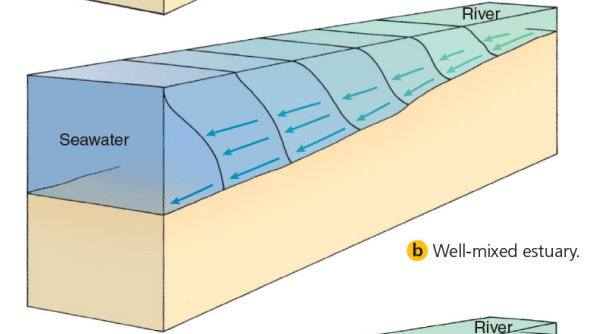


b Fjords: New Zealand's Milford Sound; the Strait of Juan de Fuca in Washington state.

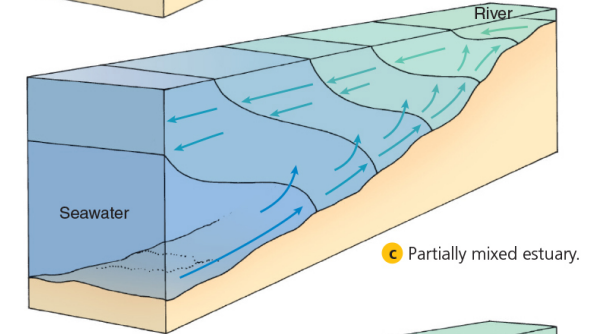
d Tectonic: San Francisco Bay; Tomales Bay (see Figure 12.11b).



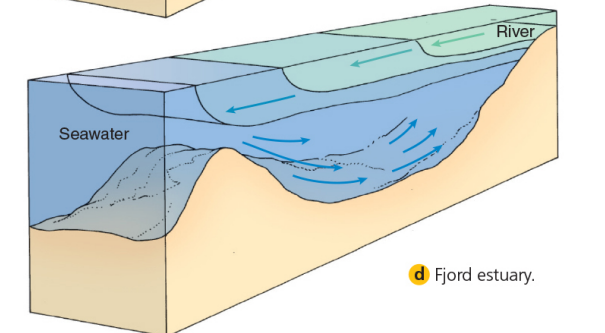
a Salt wedge estuary.



b Well-mixed estuary.



c Partially mixed estuary.



d Fjord estuary.

Figure 12.25 Estuaries classified by their origins.

Figure 12.26 Types of estuaries in vertical cross sections. The salinity values show the amount of mixing between freshwater and seawater in the various types. Green color represents freshwater.

A few examples of estuaries:

